SPECIALIZATION MAPS COMMUTE WITH FLAT PULLBACK OF RELATIVE DIMENSION E

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Specialization of cycles of integral closed subschemes flat over a DVR (relative-cycles chapter, Lemma lemma-specialization-closed, canonical tag 0H4J). cited at proof lines 534-535. Gives sp_{X'/R}([Z'_K]_{r+e}) = [Z'_kappa]_{r+e} for the flat inverse image Z'.
- Flat pullback of coherent cycles (Chow Homology, Lemma chow-lemma-pullback-coherent, canonical tag 02RE). cited at proof lines 536-538. Gives f_K^*[Z_K] = [Z'_K]_{r+e} and f_kappa^*[Z_kappa]_r = [Z'_kappa]_{r+e}.

Source reference. Original lemma 0H4J: sources/stacks-chow/tags.txt, source lines 21354–21354. Cited at proof lines 534-535: computes sp_{X'/R}([Z'_K]_{r+e}) = [Z'_kappa]_{r+e}.

Source reference. Original lemma 02RE: sources/stacks-chow/tags.txt, source lines 3621–3621. Cited at proof lines 536-538: computes both flat pullbacks f_K^*[Z_K] and f_kappa^*[Z_kappa]_r.

Lemma 0.2. *Let R be a discrete valuation ring with fraction field K and residue field κ. Let X be a scheme locally of finite type over R. Let $f : X' \to X$ be a morphism which is locally of finite type, flat, and of relative dimension e. Then the diagram*

$$\begin{array}{ccc} Z_{r+e}(X'_K) & \xrightarrow{sp_{X'/R}} & Z_{r+e}(X'_\kappa) \\ \uparrow & & \uparrow \\ Z_r(X_K) & \xrightarrow{sp_{X/R}} & Z_r(X_\kappa) \end{array}$$

commutes where $r \geq 0$ and the vertical arrows are given by flat pullback.

Proof. Let $Z \subset X$ be an integral closed subscheme dominating R. By the construction of $sp_{X/R}$ we have $sp_{X/R}([Z_K]) = [Z_\kappa]_r$ and this characterizes the specialization map. Set $Z' = f^{-1}(Z) = X' \times_X Z$. Since R is a valuation ring, Z is flat over R. Hence Z' is flat over R and $sp_{X'/R}([Z'_K]_{r+e}) = [Z'_\kappa]_{r+e}$ by Lemma 0H4J. Since by

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Chow Homology, Lemma 02RE we have $f_K^*[Z_K] = [Z'_K]_{r+e}$ and $f_\kappa^*[Z_\kappa]_r = [Z'_\kappa]_{r+e}$ we win. $\square$